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Optimizing in-band network telemetry problem

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Abstract

In-band network telemetry is a powerful framework for network monitoring. It allows the collection of telemetry data in real-time and provides network-wide visibility. However, depending on the routing of network flows and which telemetry data are collected, the network-wide visibility and the performance of monitoring applications may decrease. In this paper, we present the in-band network telemetry problem and extend the existing mathematical optimization models of the problem by proposing a new model that computes the routing of network flows. Results show that the new model outperforms existing models in term of network coverage and monitoring applications performance. The results of this work can be useful for network managers and enterprises to gain real-time insights into network performance.

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1. Introduction

Network monitoring is essential in providing high network-wide visibility in several sectors. The process consists of monitoring the network usage and performance, detecting failures, and reporting it as much as possible in real-time. The network monitoring mechanism started with adding taps in the network for collecting monitoring packets, then evolved to recording the internal states of the switches and nodes into the data packet as the packets go through the network. The last mechanism, the in-band network telemetry (INT), has recently emerged as a promising alternative to traditional network monitoring. The INT is a framework designed to monitor, collect, and report flows and network states, by the data plane, without requiring intervention or work by the control plane. Most of the recent progress in the INT has been driven by advances in programmable network devices and high level domain-specific networking descriptions and query languages [2].

Using programmable devices, in-band data plane telemetry allows us to collect and encapsulate low-level telemetry information into production network traffic. Packets contain header fields interpreted as telemetry instructions by network devices [1]. However, the INT detection of failures in the network is limited because telemetry paths and metrics must be preassigned by network operators (or control plane applications). Consequently, INT can only monitor certain telemetry data of specific paths [3]. Thus the INT coverage is most likely limited to particular devices and links in the network. Recent efforts that discuss INT orchestration include [1, 3–7]. Liu and al. [3] proposed NetVision to provide higher network-wide visibility to network operators. Their approach is based on customizing the probe path by changing the segment routing labels at run-time. However, inserting an active probe can be expensive. It may lead to queue buildups, buffer-bloat, packet drops, network congestion, and delays, mainly if performed in an unplanned ad-hoc manner [8].

In addition, the probe path must be carefully designed to achieve comprehensive network visibility. Marques et al. [5] provided the first effort towards in-band network orchestration. They proposed two heuristics showing that the network visibility may degrade in coverage, consistency, and freshness depending on which telemetry items are collected. However, the proposed heuristics do not optimally assign telemetry items to network flows due to the local decisions taken by these heuristics. Hohemberger et al. [1] formulated a mathematical model for the in-band network telemetry orchestration plan problem as a mixed integer linear programming (MILP). They proposed a machine-learning mechanism to dynamically coordinate the collection of telemetry items from the network. Recently, Hohemberger et al. [4] tackled the scalability limitation of the INT by proposing an iterated local search procedure.

Despite the effort to deliver the best out of the in-band network telemetry to improve network coverage, little has been done to optimize route flows through the network. Typically, network managers employ static routing or distance vector routing, such as the shortest path, as the primary method for routing within the network. There are several issues with these methods. First, static routing is a manual process that requires network administrators to configure and update routing tables manually. Second, manually configuring network routes becomes very challenging in large and complex networks, as keeping track of all the routes and ensuring they are up-to-date can be difficult. Third, the shortest path algorithm for routing network flows to interconnect forwarding devices does not provide full network coverage and is most likely to miss some forwarding devices within the network.

This paper proposes an extended model to the model introduced by Hohemberger et al. [1]. The new model is an automated approach for routing network flows for INT while collecting telemetry data from the network. This approach optimally assigns network flows to telemetry items and provides higher network-wide visibility with the network. The main idea involves introducing new flow variables to the model proposed by Hohemberger et al. [1]. These variables ensure more flexibility in routing network flows and allow control of the longest path length. For modeling the flow paths, the model requires the number of network flows, the maximum number of hops a path can have, and the endpoints (i.e., ingress and egress forwarding devices).

2. The in-band network telemetry problem

In-band network telemetry (INT) is a paradigm used in computer networks to collect accurate real-time telemetry data about traffic flowing through the network. It provides network-wide visibility of network traffic. INT embeds telemetry data into data packets as they travel across the network.

The in-band network telemetry problem aims to coordinate the collection of network telemetry data to maximize the efficiency of network monitoring applications by satisfying as many network monitoring requirements as possible while minimizing the impact on production network flows.

Along this paper, we will be mentioning two variations of the problem; the first variation is the one introduced by [1], which we label here as in-band network telemetry basic problem (INTB). The second variation we are proposing here is an extension to the INTB problem. We label this second variation as the in-band network telemetry extended problem (INTE).

2.1. Problem illustration

To illustrate the problem, we consider the network infrastructure of nine forwarding devices ($A, B, C, D, E, F, G, H, I$) depicted in Fig. 1., where each forwarding device embeds a subset of telemetry items $V_d \in V$. We consider four monitoring applications (M_1, M_2, M_3 , and M_4) each demanding a different subset of telemetry items R_m (i.e. monitoring requirements). We also consider one network flow packet f_1 with source node A and destination node F . The network flow f_1 has a limited capacity of collecting six telemetry items. In this example network flow packet f_1 is routed following the shortest path (3 hops) in Fig. 1a) and with a computed path of 4 hops in Fig. 1b). From Fig. 1b), the requirements of the four monitoring application are satisfied by routing the network flow packet f_1 from A to F . The requirements of monitoring applications M_1 have been satisfied by collecting the yellow and green items from the same forwarding device in (2). The requirement of monitoring application M_3 also is satisfied by collecting both the yellow and blue items from the same forwarding device in (3). The monitoring requirements of monitoring applications M_2 and M_4 are satisfied by collecting the red item from forwarding device E in (4) and the green item from forwarding device F in (5).

In Fig. 1a), the requirement of monitoring applications M_1 is satisfied as both yellow and green items have been collected from the forwarding device G by the packet in (3). The requirement of monitoring application M_4 has been satisfied in (5) as the green item is collected. The requirement of monitoring application M_2 has been satisfied in (2). However, the packet failed to meet the requirements of monitoring application M_4 at devices A and H , which demanded both the blue and yellow items to be collected from the same device.

The solution in Fig. 1b) with the longer route is optimal and satisfies the maximum number of monitoring requirements since it has satisfied the monitoring requirements of all four monitoring applications. The solution in Fig. 1a) with a shorter route failed in satisfying the monitoring requirement of monitoring application M_3 because blue and yellow items were not collected from the forwarding device.

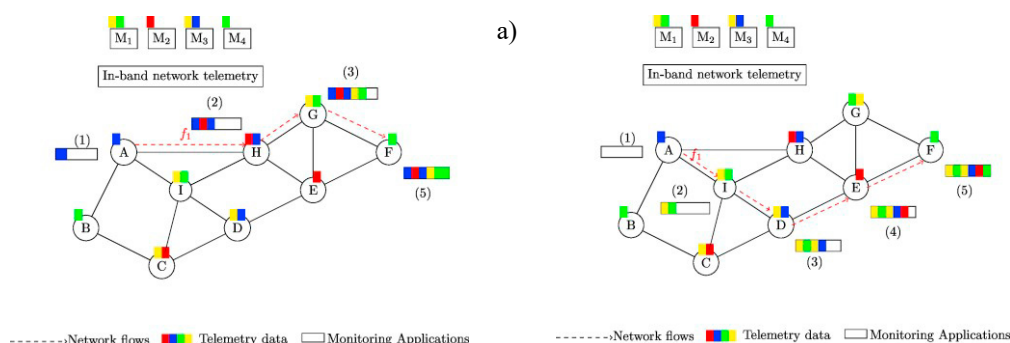


Fig. 1. Illustration of the in-band network telemetry problem and the impact of network flow routing on the solution. (a) shortest path; (b) computed path.

2.2. INTE mathematical model

This section presents an extended problem of the INTB problem by introducing a new set of variables of flows that allows some active network flows paths to be computed. In this problem, some active network flow paths can be given while others are calculated within the model. The mathematical model inputs include:

- a physical network infrastructure $G = (D, L)$, where D is the set of forwarding devices and L is the set of links
- V_d , set of telemetry items embedded by each device d .
- $\delta(v)$, size of each telemetry item $v \in V_d$.
- a set of active network flows F and their capacity K_f .
- percentage of active network flows to be given with pre-defined routes GF . model will compute the remaining routes of active network flows.
- max_route , is an integer limiting the maximum number of hops in a route.
- two endpoints S_f and E_f for each active network flow.

The outputs for the INTE model are denoted by the 4-tuple $X = \{X, Y, S, T\}$ where,

- $Y = \{y_{dvf}, \forall d \in D, v \in V, f \in F\}$ is the set of collected telemetry items.
- $S = \{s_{mdp}, \forall m \in M, d \in D, p \in P\}$ is the set of satisfied spatial dependencies.
- $T = \{t_{mp}, \forall m \in M, p \in P\}$ is the set of satisfied temporal dependencies.
- $X = \{x_{ijf}, \forall i \in D, j \in D, f \in F \setminus GF\}$ is the set of routes for non-defined routes of active network flows.

$$\text{Maximize } \sum_{m \in M} \sum_{d \in D} \sum_{p \in P \in R_m^s} s_{mdp} + \sum_{m \in M} \sum_{p \in P \in R_m^t} t_{mp} \quad (1)$$

Subject to:

$$\sum_{d \in P(f)} \sum_{v \in V_d} \delta(v) \cdot y_{dvf} \leq K(f) \quad \forall f \in GF \quad (2)$$

$$\sum_{j \in N^+(i)} x_{ijf} = 1 \quad i = S_f, \forall f \in F, GF \quad (3)$$

$$\sum_{i \in N^-(j)} x_{ijf} = 1 \quad j = E_f, \forall f \in F, GF \quad (4)$$

$$\sum_{j \in N(i)} x_{jif} - \sum_{j \in N(i)} x_{ijf} = 0 \quad \forall i \in D, \{S_f, E_f\}, f \in F, GF \quad (5)$$

$$\sum_{j \in N^+(i)} x_{ijf} \leq 1 \quad \forall i \in D, \{S_f, E_f\}, f \in F, GF \quad (6)$$

$$\sum_{(i,j) \in L} x_{ijf} \leq max_route \quad \forall f \in F, GF \quad (7)$$

$$y_{dvf} \leq \sum_{j \in N^+(d)} x_{djf} \quad \forall d \in D, v \in V_d, f \in F, GF \quad (8)$$

$$\sum_{f \in F} y_{dvf} \leq 1 \quad \forall d \in D, v \in V_d \quad (9)$$

$$\sum_{d \in D} \sum_{v \in V_d} \delta(v) \cdot y_{dvf} \leq K(f) \quad \forall f \in F, GF \quad (10)$$

$$a_{mdp} = \sum_{v \in P} \sum_{f \in F} y_{dvf} \quad \forall m \in M, p \in P \in R_m^s, d \in D \quad (11)$$

$$s_{mdp} \leq \frac{a_{mdp}}{|P|} \quad \forall m \in M, p \in P \in R_m^s, d \in D \quad (12)$$

$$b_{mp} = \sum_{d \in D} \sum_{v \in P} \sum_{f \in F} y_{dvf} \quad \forall m \in M, p \in P \in R_m^t : \zeta(P) > \xi(P) \quad (13)$$

$$t_{mp} \leq \frac{b_{mp}}{|P|} \quad \forall m \in M, p \in P \in R_m^t \quad (14)$$

$$x_{ijf} \in \{0,1\} \quad \forall (i,j) \in L, f \in F \quad (15)$$

$$y_{dvf} \in \{0,1\} \quad \forall d \in D, v \in V_d, f \in F \quad (16)$$

$$s_{mdp} \in \{0,1\} \quad \forall m \in M, d \in D, p \in P \in R_m^s \quad (17)$$

$$t_{mp} \in \{0,1\} \quad \forall m \in M, p \in P \in R_m^t \quad (18)$$

$$a_{mdp} \in \{0,1\} \quad \forall m \in M, d \in D, p \in P \in R_m^s \quad (19)$$

$$b_{mp} \in \{0,1\} \quad \forall m \in M, p \in P \in R_m^t \quad (20)$$

The objective function (1) is to maximize the number of spatial and temporal dependencies. Constraints (2) ensure that no active network flow with a pre-defined path exceeds its capacity. Constraints (3) and (4) ensure that each flow will start at its origin and end at its destination. Constraints (5) are the flow conservation constraint (i.e. for every node d in the path of flow $f \in F$ different from the origin and destination of the flow, there is precisely one arc entering d and one arc leaving d). Constraints (6) are for sub-tour elimination, and constraint (7) limit the maximum route a flow can have; the *max_route* is an upper limit to the total number of hops an active network flow can achieve. Constraints (8) ensure that a telemetry item v is collected by an active network flow only if the device d that collects it belongs to the path of the active network flow f . Constraints (9) ensure that a single telemetry item is collected at most by one network flow $f \in F$ in a given device d . Constraints (10) ensure that no flow $f \in F$ exceeds its capacity. Constraints (11) and (12) aim to count whether or not a spatial dependency is met. Given a spatial dependency, constraints (11) counts the number of telemetry items collected from a device d . Constraints (12) verify whether or not the dependency is met (i.e., checking if all items were collected). Similarly, constraints (13) and (14) count and verify the number of satisfied temporal dependencies. Last, the constraints (15)-(20) define the variables' output domains.

3. Computational experiments

This section presents the computational experiments with the mathematical model introduced in the previous section. We first describe the experimental setup and the dataset used, then we evaluate three sets of experiments. First, we assess the network-wide visibility regarding network telemetry coverage (i.e., the percentage of telemetry items collected) concerning an increasing number of active network flows with up to 40% of the active network flows with a non-pre-defined route computed while solving the optimization model. Then, we evaluate the monitoring applications' performance (i.e. in terms of the number of satisfied monitoring requirements), considering an increasing number of active network flows and monitoring applications. Last, we compare our results to Hohemberger et al. [1] optimization problem.

3.1. Setup

All the experiments presented in this paper were conducted on a machine with forty-seven Intel(R) Xeon(R) E5-2697 processors running at 2.70 GHz and with 64GB of RAM, using Ubuntu GNU/Linux Server 18.04.5 LTS x86_64 operating system. The mathematical model was coded and implemented in Python 3.7 using the Docplex package and solved by Cplex 12.10 under an academic license and using the default parameter values. The maximum computation time limit to find a solution was set to 14,400 seconds (4h).

3.2. Dataset

The dataset used in the experiments to evaluate the in-band network telemetry problem comprises a physical network infrastructure generated following the Barabasi-Albert model [9]. We consider a physical network infrastructure of 50 nodes (forwarding devices) and an average of 200 physical links. On top of each network infrastructure instance, we considered a set F of network flows and varied the size of $|F|$ from 50 to 200. Each network flow has a limited capacity K_f ranging from 2-20 bytes. Forwarding devices have 8 possible telemetry items of size $\delta(v)$ varying from 2-10 bytes to export [6].

We generated multiple instances of network infrastructure and named each instance $DxVyMz$. For example, the instance $D50T8M2$ corresponds to an instance of a network infrastructure of 50 forwarding devices, each forwarding device embedding 8 telemetry items and 2 monitoring applications are used.

3.3. Results

This section presents the results for two models of the in-band network telemetry extended problem. The first model, INTE-80%, considers a network infrastructure where 80% of network flows are given as input and routed following the shortest path. The second model, INTE-60% considers a network infrastructure where 60% of network flows are given as input and routed following the shortest path. The models determine the routes of the remaining network flows. The results of the experiments are summarized in Table 1. We base our evaluation of the results on two aspects of network monitoring: the network wide-visibility, the monitoring requirement satisfaction and network flow routing.

The network wide-visibility is evaluated by the number of telemetry items collected for the network (the column “Items” from Table 1.). We observe that, the number of collected telemetry items with both models INTE-80% and INTE-60% increases as the number of network flows does. We also noticed that for all instances, the INTE-60% model collected more telemetry items, thus it gave more network-wide visibility. This can be explained by the fact that some of the network flows have multiple possible routes from the source to the destination; hence, these network flows can visit and collect telemetry data from more forwarding devices in the network.

The column “Objective” from Table 1., which corresponds to the objective function shows that the number of satisfied monitoring requirements increases for each instance as the number of monitoring applications and network flows increase.

Table 1. Comparative Analysis of the optimization models considering different network infrastructures with an increasing number of monitoring applications and growing number of network flows.

Instance	Flow	Items	Objective	INTE-80%			Items	Objective	INTE-60%		
				UB	Gap(%)	CPU(s)			UB	Gap(%)	CPU(s)
D50V8M2	50	113	696*	696	0	7960.91	123	740	752	1.64	14400
	100	233	1378	1398	1.51	14400	242	1468	1530	4.22	14400
	150	250	1530*	1530	0	92.53	250	1530*	1530	0	794.44
	200	250	1530*	1530	0	30.4	250	1530*	1530	0	365.16
Average		211.5	1283.5	1288.5	0.38	5620.96	216.25	1317	1335.5	1.47	7489.9
D50V8M4	50	121	864*	864	0	8051.44	141	936	959	2.47	14400
	100	233	1579	1602	1.5	14400	259	1767	1836	3.91	14400
	150	318	2143	2187	2.08	14400	343	2272	2365	4.1	14400
	200	385	2418	2428	0.43	14400	400	2448*	2448	0	14400
Average		264.25	1751	1770.25	1	12812.86	285.75	1855.75	1902	2.62	14400
D50V8M6	50	126	977	981	0.44	14400	144	1057	1084	2.59	14400
	100	231	1750	1779	1.68	14400	260	1968	2015	2.41	14400
	150	316	2334	2346	0.56	14400	336	2402	2541	5.81	14400
	200	383	2686	2712	1	14400	399	2746	2754	0.29	14400
Average		264	1936.75	1954.5	0.92	14400	284.75	2043.25	2098.5	2.78	14400
D50V8M8	50	110	957	961	0.49	14400	130	1064	1092	2.65	14400
	100	214	1753	1775	1.29	14400	241	1994	2042	2.42	14400
	150	309	2582	2638	2.21	14400	344	2900	2977	2.69	14400

	200	378	3074	3105	1.01	14400	391	3112	3162	1.61	14400
	Average	252.75	2091.5	2119.75	1.25	14400	276.5	2267.5	2318.25	2.34	14400
D50V8M10	50	113	1075	1104	2.71	14400	131	1227	1274	3.9	14400
	100	215	2038	2083	2.24	14400	248	2227	2380	6.88	14400
	150	305	2989	3050	2.05	14400	342	3242	3472	7.13	14400
	200	380	3614	3693	2.19	14400	400	3774*	3774	0	14400
	Average	253.25	2429	2482.5	2.3	14400	280.25	2617.5	2725	4.48	14400
Overall average		249.15	1913.35	1923.1	1.17	12326.76	268.7	2020.2	2075.85	2.74	13017.98

To evaluate the impact of the network flows routing on the satisfied monitoring requirements, we ran the same experiments with the INTB model. We compared the results with the INTE-80% and INTE-60% models. Fig 2. shows that for an increasing number of monitoring applications, the best average of satisfied monitoring applications from each instance was obtained by the INTE-60% model. The flexibility in the network flow routes can explain this.

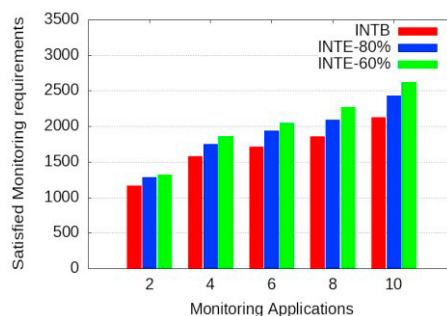


Fig. 2. Comparison of monitoring applications' performance.

4. Conclusion

In this paper, we proposed a new variation of the in-band network telemetry problem that has more flexibility over network flow routing. We implemented it and solved it using the IBM Cplex solver. We showed the impact of the network flows routing on the network-wide visibility and performance of the monitoring applications. We also showed that the solution of our mathematical model provides a network-wide visibility and high performance of network monitoring applications compared to the state-of-the-art optimization models. The results of this work provide substantial advantages for network managers and enterprises. Through network-wide visibility and monitoring application performance, it enables them to monitor network performance in real-time, enabling proactive decision-making, optimizing resource allocation, and boosting network security and dependability.

As future work, it is proposed to implement a heuristic approach for our model to test bigger networks and reduce the computational time. Furthermore, it is proposed to test the model on different types of network infrastructures.

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